Dynamic-Hierarchical Attention Distillation With Synergetic Instance Selection for Land Cover Classification Using Missing Heterogeneity Images

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***Abstract*— Optical and SAR modalities can provide the complementary information on the land properties, which usually lead to more robust and better classification performance. However, due to the restriction of imaging condition, not all modalities included into the training data sets could be available in real testing samples. Therefore, it is important to explore how to learn discriminative representations using multimodal data during the training stage, while achieving fine land cover classification using missing modalities at test time. In this article, we propose a novel dynamic-hierarchical attention distillation network (DH-ADNet) with multimodal synergetic instance selection (MSIS) for land cover classification using missing data modalities. First, the MSIS realizes the selection of the most representative multimodal instances to enhance the DH-ADNet's ability of discriminative feature extraction. Then, the DH-ADNet is training on the basis of the curriculum learning strategy and promotes the hallucination stream to learn the privileged information. In particular, a novel dynamic-hierarchical attention distillation module (DH-ADM) is introduced, which adaptively highlights different contributions of multilayer attention distillation by carefully exploring the classification losses of multilayer features over the training iterations. Comprehensive evaluations on two coregistered optical and SAR data sets and report state-of-the-art results in the privileged information scenario.**

***Index Terms*— Convolutional neural networks, distillation loss, land cover classification, missing modalities, privileged information.**

I. Introduction

IN RECENT years, the improvements of remote sensing (RS) image processing technologies based on multiple modalities have provided new opportunities for more accurate land cover classification [1]–[5], since they integrate the complementary information from different spatial resolution, spectral bands, and imaging modes, which have been widely considered into the environmental monitoring [1], agricultural resources statistics [2], [3], and urban local climate zone scheme establishment [5], [6]. However, multiple modalities-based actual operations for high quality and real-time land cover maps will encounter the problem of missing modal [7], [8], because there are certain restrictions: 1) due to the satellites have longer revisit period, it is impossible to obtain RS images from the same satellite of the same areas in time. For instance, Copernicus Sentinel-2 developed by the European Space Agency (ESA) has a revisit period of five days and 2) climate conditions limit the acquisition of high-quality optical RS images from the areas because the clouds and rain will drastically affect the measured electromagnetic waves, however, SAR images have all-time and all-weather imaging characteristics.

Previous mostly multimodal-based land cover classification approaches have a premise that all modalities are complete and are not missing during training and test times [9]. Therefore, when the modalities are missing during test time, it is a challenging question that how to use all data available to improve the land cover classification. Fortunately, pattern recognition algorithms with missing data have a certain research foundation [10]–[13] and can be grouped into four main groups.

1) Incomplete modalities are deleted while complete modalities are reserved and used to complete the feature extraction and classifier design. However, the category methods will be lost the large amounts of potentially information that can improve the performance of pattern recognition.
2) According to the potential correlation between different modalities, the incomplete modalities are imputed based on the observed elements. However, imputation methods can get better results under the missing at random (MAR) [11].
3) Model-based approaches for handling missing data that can model the distribution of existing data modalities [12].
4) The missing modalities are embedded into machine learning approaches and are disposed of [13], 14], e.g., decision trees and support vector machines.

Some pattern recognition algorithms for handling partly missing data have achieved good experimental results, but they do not work well when the modality is completely missing. Meanwhile, there is a semantic gap between optical and SAR images, it is difficult for the shallow features of classics machine learning algorithms to form a unified description. Furthermore, as the resolution increases of RS data, the feature dimension of high-resolution RS images is increasing, which is

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harmful to the above statistic algorithms on handling missing data.

Recently, deep neural networks as a type of multilayer and end-to-end trainable models that are capable of capturing hierarchical discriminative features, which have been successfully applied to computer vision, image classification [15], [16], segmentation [17], object detection [18], and land cover classification with complete modality [49]. Numerous review articles on land cover classification based on deep learning have not only pointed out the challenges of land cover classification based on high-resolution RS images but also have confirmed the superiority of deep learning in land cover classification based on high-resolution RS images [49], [50]. Liu *et al*. [51] proposed a novel dense dilated convolutions merging network, which merged the feature map of each layer with the feature maps from all previous layers to obtain multiscale information, and the experiments have demonstrated its effectiveness. Meanwhile, deep learning algorithms also have given rise to opportunities on land cover classification with completely missing data sources. For instance, a series of missing modalities imputation methods based on deep learning, such as cascaded residual auto-encoder [19], deep learning-based forecasting ensemble [20], and missing modalities imputation frameworks based on generative adversarial networks [21] have been proposed to compound missing modalities based on existing modalities. However, the missing modalities imputation methods based on deep learning can be considered as data postprocessing in pattern recognition tasks, which will greatly influence the follow-up work of pattern recognition.

More recently, the main target of learning using privileged information or learning with side information is to be able to utilize extra knowledge only available during the training phase [22], which has also been studied recently in the context of deep learning to handle missing data [22], [23]. Modality hallucination network and deep multitask learning are two straightforward approaches that can incorporate the privileged information [7], [22], [24], [25]. Modality hallucination network models are used to mimic the missing modality based on the distillation loss [22], [24]. Early knowledge distillation methods were usually to distill the knowledge from the last layers of the student and teacher networks [24], [26]–[28]. And then for further increasing the accuracy of the student network, the multilayer distillation methods were proposed [29]–[32]. However, these methods typically ignored that the multilayer knowledge distillation requires different contributions for each one during different learning phases.

Meanwhile, deep learning-based land cover classification methods as the data-driven models, the quality of input samples will greatly affect the performance of land cover classification. Generally, the randomly selection of training samples entered into the deep convolutional neural networks (DCNNs) is a common strategy, which may introduce some easy-confused samples and cause low discriminative power of the feature [33]. Therefore, before training DCNNs, the selection of representative multimodal instances plays an important role.

To address these issues, in this article, we propose a novel dynamic-hierarchical attention distillation network (DH-ADNet) with multimodal synergetic instance selection (MSIS) for land cover classification using missing data modalities. Taking the missing SAR modality as an example, the overall architecture of our DH-ADNet is shown in Fig. 1. First, the MSIS is used to choose which instances from the training set should be retained and enhance the DH-ADNet's ability of discriminative feature extraction. Then, the DH-ADNet is proposed for land cover classification with missing data modalities, which consists of two category subnetworks: the modality-specific streams (optical and SAR streams) and the hallucination stream, and the pretrained AlexNet without fully connected layer is used as the backbone. Where the optical and SAR streams are used to learn modality-specific features, respectively, and the hallucination stream learns the privileged information by mimicking the feature distribution of SAR stream in Fig. 1(b), respectively. In particular, to improve the learning ability of privileged information for the hallucination stream, the dynamic-hierarchical attention distillation module (DH-ADM) is embedded between different feature layers of the hallucination and modality-specific streams in Fig. 1(d), respectively. Finally, inspired by Jin *et al.* [34], we consider the network training from the perspective of curriculum learning [35]. Comprehensive evaluations on two coregistered optical and SAR data sets demonstrate the effectiveness of the proposed method.

The major contributions of this work are summarized threefold.

1) We propose a novel DH-ADNet to exploit the optical and SAR data at training while using only one at test time. In particular, the DH-ADM is introduced, which not only is capable of performing multilayer distillation between the SAR and hallucination streams, but also adaptive highlighting different contributions of multilayer distillation.
2) We propose the MSIS to realize the selection of most representative multimodal instances and enhance the DH-ADNet's ability of discriminative feature extraction.
3) Extensive experiments and ablation studies are performed to examining the effectiveness of the proposed DH-ADNet, in which state-of-the-art land cover classification results in the privileged information scenario on two broad coregistered optical and SAR data sets.

II. Related Work

A. Instance Selection

The aim of instance selection is to select representative instances and then structure the training data set [36]. Czarnowski [36] proposed a cluster-based instance selection approach and discusses its four variants, meanwhile the comprehensive experiment has proven that the proposed method was able to cluster training data into different groups. Instance selection is regarded as the preprocessing step of the machine learning algorithms, which can enhance the robustness and generalization properties of machine learning algorithms, and improve the discrimination of features [37]–[39]. Where

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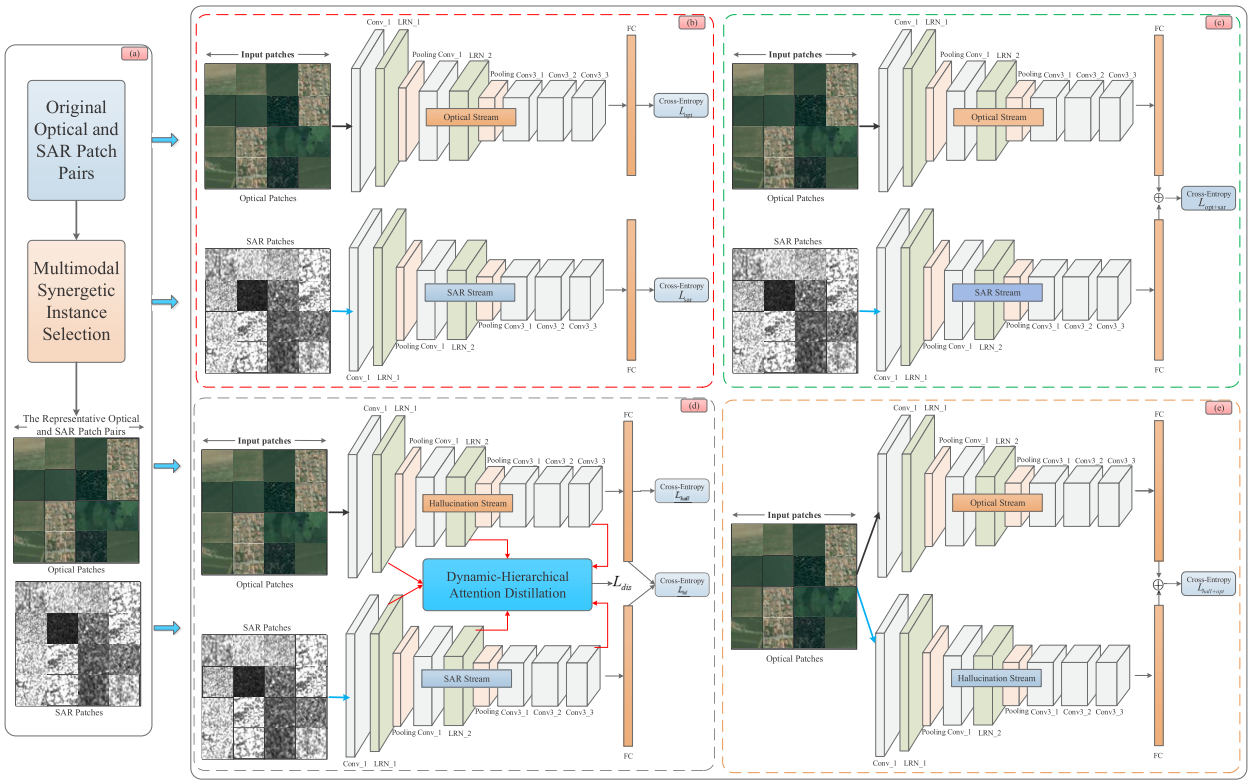


Fig. 1. Overall architecture of our DH-ADNet with missing SAR modality as an example, where (a) presents the process of the MSIS, (b) and (c) present the training of the teacher network and (b) and (c) present the training of the student network. Meanwhile, the overall architecture of our DH-ADNet with missing optical modality is obtained, when 'hallucination stream' and 'SAR stream' are modified into 'Optical stream' and 'hallucination stream' in (d), and the 'Optical stream' is modified into 'SAR stream' and the input samples are SAR patches in (e). Our DH-ADNet consists of two parts: 1) MSIS realizes the selection of most representative multimodal instances and train our DH-ADNet for enhancing the discrimination of the learned features and 2) DH-ADM can perform multilayer distillation between SAR and hallucination streams and adaptive highlighting different contributions of multilayer distillation. $\oplus$ is the element-addition.

distance-based instance selection algorithms are used to a type of common instance selection algorithms, which select representative instances based on the calculation of the distance between prototypes and training samples [38]. Wilson and Martinez [38] proposed a novel integrated decremental instance-based learning algorithm (IDIBL) for overcoming the weaknesses of instance selection based on basic nearest-neighbor, which was able to reduce storage and improve generalization accuracy. Meanwhile, other groups of representative methods are the family of IRA1-IRA4 [39] and clustering methods [40], [41], which attempted to delete unnecessary training examples based on artificial criteria and structure of the compact training data set. Anderson *et al*. [41] studied the pros and cons of five clustering-based training set selection methods on the quantitative chemical analysis of geologic samples, the comprehensive experiment has proven that the clustering-based training set selection methods have a certain effect on the improvement of the method. However, the above methods are suitable for single-mode instance selection.

In this article, we propose the MSIS based on the cluster method, the collaborative of which consider optical and SAR to realize the representative instances selection of optical and SAR modals, simultaneously.

B. Generalized Distillation

The knowledge distillation introduced by Hinton *et al.* [42] and the learning of privileged information introduced by Vapnik *et al.* [21] as two instances of the same learning paradigm [23], which is termed generalized distillation. Where the generalized distillation methods introduced the intelligent teacher that supplied additional information to guide the learning of the student network, so that the "student" network can use these additional explanations proposed by the "teacher" network at test time and outperform those methods based on the single modal or features.

Many scholars are devoted to study how to improve the performance of distillation. Early knowledge distillation methods were usually to distill the knowledge from the last layers of the student and teacher networks [7], [42], [43]. Hinton *et al.* [42] proposed simple but elegance methods that raised the temperature of the final softmax to produce a suitably soft set of targets and realized the distillation between student and teacher networks. Tang *et al.* [43] took a simple deep neural network model as the teacher network, which was used to train the student network, the complex recurrent neural network, and the experiments on automatic speech recognition (ASR) task showed that the complex recurrent neural network was able to obtain good performance. After that, multilayer knowledge distillation methods have been proposed to further increasing the accuracy of the student network [32], [44]. Romero *et al.* [32] realized that parameters from middle layers of teacher and student networks were able to improve the distillation performance, therefore, the proposed FitNet treated the output representations and the intermediate representations from the teacher network as hints to improve the distillation performance of the student network. Moreover, the attention

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TABLE I

DETAILED NETWORK ARCHITECTURES OF EACH STREAM. ALEXNET* DENOTES THE ORIGINAL ALEXNET WITHOUT THE CLASSIFIER, AND THE STRIDE OF CONV_1 LAYER IS MODIFIED TO 1. MEANWHILE, THE ALEXNET* IS PERTRAINED BASED ON IMAGENET

Name	Structure	Stride	Padding	Component
Conv_1+ReLU	11×11,96	1	5	Encoder based on AlexNet*
Local Response Norm(LRN_1)				
MaxPooling	3×3	2	1	
Conv_2+ReLU	5×5, 256	1	2	
Local Response Norm (LRN_2)				
MaxPooling	3×3	2	1	
Conv3_1+ReLU	3×3, 384	1	1	
Conv3_2+ReLU	3×3, 384	1	1	
Conv3_3+ReLU	3×3, 256	1	1	
Fully Connected Layer (FC)				Classifier
Output	Softmax			

information has been used to improve the performance of the student network based on the distillation between the attention maps from a weakly student network and a powerful teacher network [31], [45]. However, the above methods ignored that multilayer knowledge distillation requires different contributions when neural networks undergo different learning phases during the training process.

In this context, the closet works to our proposed method are [7] and [31], Differently, we propose a novel DH-ADM, which not only uses the attention distillation distance between different layer features of the hallucination and SAR streams, but also dynamics adjust the contribution of different layers of distillation features.

III. METHODOLOGY

In this section, we present a novel DH-ADNet with MSIS for land cover classification with missing modalities. Taking the land cover classification with missing SAR as an example, Fig. 1 illustrates the overview of our method. First, we propose the use of MSIS to choose which instances from the training set should be retained during the learning process. Then, the DH-ADNet is proposed, where the modality-specific streams (the optical and SAR streams) learn modality-specific features, respectively, and the hallucination stream learns the privileged information by mimicking the feature distribution of the SAR stream, the pretrained AlexNet without fully connected layers as the encoder, the detailed network architectures of each stream are described in Table I. Similarly, the overall architecture of our DH-ADNet with missing optical modality is obtained, when "hallucination stream" and "SAR stream" are modified into "Optical stream" and "hallucination stream" in Fig. 1(d), and the "Optical stream" is modified into "SAR stream" and the input samples are SAR patches in Fig. 1(e). In particular, to improve the privileged information learning ability of hallucination stream, the DH-ADM is embedded between different layer feature between the hallucination and SAR streams (here the features of the *LRN_1*, *LRN_2*, and *Conv3_3* are distilled. Finally, inspired by the work [34], we consider the network training from the perspective of curriculum learning.

A. Multimodal Synergetic Instance Selection

The DH-ADNet as a data-driven model, the quality of the training samples will greatly affect the performance of the model. Generally, the training samples entering the DCNNs are selected at random, which may cause the training sample to contain some easy-confused samples, and improve the discriminative of the learning feature [33]. Therefore, we propose a novel MSIS method to realize the selection of the most representative multimodal instances and train the DH-ADNet for enhancing the discrimination of the learned features. The overall processing of the MSIS method is shown in Algorithm 1, and the details are as follows.

1) We randomly pick $L = 10\,000$ optical and SAR patch pairs $Z^k = \{z_i^k = \{x_i^k, y_i^k\}\}_i^L$ of the kth land cover type from the raw optical and corresponding SAR patches. We use the K-means algorithm to respectively cluster the optical and SAR patches, and obtain the centers $\{\tilde{u}_i^k\}_i^M$ and $\{\hat{u}_i^k\}_j^M$ of optical and SAR patches, where M is the number of cluster center.
2) We realize the alignment between the optical and SAR cluster centers $\{\tilde{u}_i^k\}_i^M$ and $\{\hat{u}_j^k\}_j^M$ based on the Euclidean distance as follows:

$$d_{ij}^k = \left\|\tilde{u}_i^k - \hat{u}_j^k\right\|_2 \tag{1}$$

where $\|\cdot\|_2$ denotes the ℓ_2-norm, and the distance matrix $D^k = \{d_{ij}^k | i, j \in \{1, \ldots, M\} \& i \neq j\}$.The distance vector D^k is sorted with ascending order, and according to the index of the element in the sorted D^k, we can obtain the alignment between optical and SAR cluster centers. For example, when the index of the d_{12}^k is 12, which denote the 1th optical cluster centers and 2th SAR cluster centers are aligned. Therefore, we obtain the aligned cluster center set $Q^k = \{(\tilde{u}_i^k, \hat{u}_j^k)\}_{i,j}^M$ of the kth land cover type.
Similarly, we also obtain the alignment of aligned optical and SAR cluster centers $\{Q^1, \ldots, Q^K\}$ for different land cover type, where K is the number of land cover types.
3) We obtain the correlation between the cluster centers of different land cover types based on the cosine distance, and then the top $M/2$ discriminative prototypes of each land cover type are selected, as follows:

$$\text{Corr}_j^k = \frac{1}{KM}\sum_{i=1}^{K}\sum_{m=1}^{M}\frac{\left(q_j^k\right)^T q_m^i}{\left\|q_j^k\right\|_2\left\|q_m^i\right\|_2}, \quad \text{s.t. } i \neq k \tag{2}$$

where $q_j^k = (\tilde{u}_j^k, \hat{u}_j^k)$ denotes the jth aligned vector in the Q^k for the kth land cover type, and $q_m^i = (\tilde{u}_m^i, \hat{u}_m^i)$ denotes the mth aligned vector of the ith land cover type.Then, the correlation vector Corr^k is sorted with ascending order, where the first $M/2$ element is selected as the discriminative prototype of the kth land cover types $\tilde{Q}^k = \{\tilde{q}_1^k, \ldots, \tilde{q}_{M/2}^k\}$.
Similarly, we also obtain $M/2$ discriminative prototypes $\{\tilde{Q}^1, \ldots, \tilde{Q}^K\}$ for all land cover type.
4) The optical and SAR patches of each land cover type near the corresponding prototypes are defined as representative patches, while the distant ones are defined as easy-confused patches

$$\tilde{d}_{ij}^k = \lambda\left\|\tilde{x}_i^k - \tilde{u}_j^k\right\|_2 + (1-\lambda)\left\|\tilde{y}_i^k - \hat{u}_j^k\right\|_2 \tag{3}$$

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Algorithm 1 Overall processing of MSIS

Input: Randomly pick the optical and SAR patch pairs Z^k;
Output: The representative optical and SAR patches $\tilde{Z}^k$;

1: Cluster the optical and SAR patches Z^k,and obtain the center sets $\{\tilde{u}_i^k\}_i^M$ and $\{\hat{u}_j^k\}_j^M$;

2: Align the optical and SAR cluster centers $\{\tilde{u}_i^k\}_i^M$ and $\{\hat{u}_j^k\}_j^M$, and obtain the alignment aligned optical and SAR cluster centers $\{Q^1, \cdots, Q^K\}$ for different land cover type;

3: Select top $M/2$ discriminative prototypes $\{\tilde{Q}^1, \cdots, \tilde{Q}^K\}$ for each land cover type by (2);

4: according to (3), select the representative optical and SAR patches $\tilde{Z}^k = \{\tilde{z}_i^k = \{\tilde{x}_i^k, \tilde{y}_i^k\}\}_i^N$ for each land cover type based on the corresponding prototypes $\{\tilde{Q}^1, \cdots, \tilde{Q}^K\}$

where $\{\tilde{x}_i^k, \tilde{y}_i^k\}$ denote the ith of the kth land cover type, the $\tilde{d}_{ij}^k$ denotes the distance between $\{\tilde{x}_i^k, \tilde{y}_i^k\}$ and ith prototype of the kth land cover type. When $\tilde{d}_{ij}^k \leq \delta$, $\{\tilde{x}_i^k, \tilde{y}_i^k\}$ is selected as the training sample, and λ is balance parameter, which is used to achieve the collaborative selection of multimodal training samples. Finally, we can obtain the $\tilde{Z}^k = \{\tilde{z}_i^k = \{\tilde{x}_i^k, \tilde{y}_i^k\}\}_i^N$ representative optical and SAR patches of each land cover type.

B. Dynamic-Hierarchical Attention Distillation

Distillation of the hallucination network (called distillation of the student network) is an important component of the DH-ADNet, which mimics the information flow paths of the SAR network in Fig. 1(d). The distillation performance of the hallucination network largely determines the land cover classification effect of the DH-ADNet within missing data modalities. Early multilayer knowledge distillation methods typically ignored that the multilayer knowledge distillation requires different contributions for each one when neural networks during different learning phases. Therefore, we propose a novel knowledge distillation method, called the DH-ADM (shown in Fig. 2), which dynamic models the different contribution weights of various layers distillation between the hallucination and SAR stream based on the classification losses of multilayer features over the training iterations in Fig. 1(d), and then train a hallucination network to mimic the information flow of the SAR stream.

The DH-ADM is embedded into the *LRN_1*, *LRN_2*, and *Conv3_3* layers of backbone in Fig. 1(d). Take the convolutional feature maps $\tilde{X}_i$ and $\tilde{Y}_i$ from ith layer of hallucination and SAR streams as an example, the overall processing of the DH-ADM is described as follows.

1) We respectively obtain the spatial attention map F_i^x of the features $\tilde{X}_i$ and as follows:

$$F_i^x = \sum_{i=1}^{C} \left|\mathrm{Conv}\left(\tilde{X}_i\right)\right|^p \tag{4}$$

where $\mathrm{Conv}(\cdot)$ denotes the convolutional operation with 1×1 filter, $|\cdot|$ is the absolute value operation, $C$is the channel number of $\tilde{X}_i$.

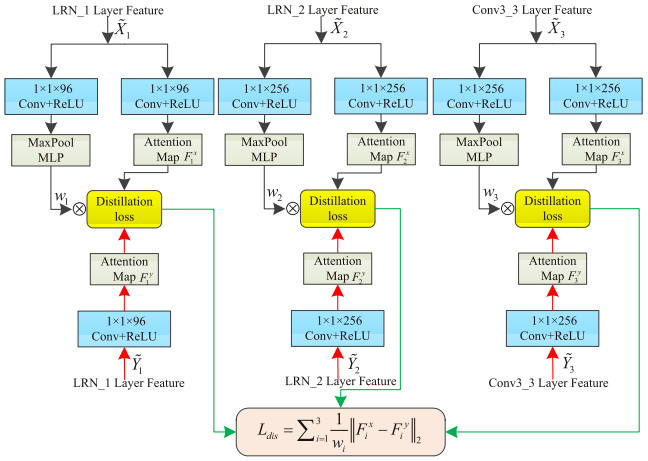


Fig. 2. Flowchart of the proposed DH-ADM. ⊗ denotes the multiplication.

Similarly, we can calculate the spatial attention map F_i^y of the $\tilde{Y}_i$. Therefore, the spatial attention maps of the LRN_1, LRN_2 and Conv3_3 layers are the $\{F_1^x, F_1^y\}$, $\{F_2^x, F_2^y\}$ and $\{F_3^x, F_3^y\}$, respectively.

2) To measure the importance of different layer features for information distillation, we consider the classification loss magnitude of different layers (*LRN_1*, *LRN_2* and *Conv3_3*) to be the measure of importance for multilayer distillations and accordingly decide on the weight for the distillation losses. We can obtain the weight value of distillation loss for i_{th} layer of hallucination and SAR streams as follows:

$$f_i = \mathrm{Mlp}\left(\mathrm{MaxPool}\left(\mathrm{Conv}\left(\tilde{X}_i\right)\right)\right) \tag{5}$$

$$w_i = -\frac{1}{N}\sum_{n=1}^{N} l_n \log\left(f_i^n\right) \tag{6}$$

where f_i^n denotes the n_{th} feature of f_i, l_n denotes the true label of sample n, the $\mathrm{MaxPool}(\cdot)$ is the global maximum pooling and $\mathrm{Mlp}(\cdot)$ denotes the multilayer perceptron with a hidden layer.

Similarly, we can calculate the weight values of the *LRN_1*, *LRN_2* and *Conv3_3* layers are w_1, w_2 and w_3, respectively.

3) We calculate dynamic-hierarchical attention distillation loss based on the Euclidean distance as follows:

$$L_{\mathrm{dis}} = \sum_{i=1}^{3} \frac{1}{w_i}\left\|F_i^x - F_i^y\right\|_2^2 \tag{7}$$

where the classification loss magnitude w_i is less, it signifies that the distillation of this layer is important, and the corresponding weight may be increased. On the other hand, a larger classification loss magnitude w_i suggests reducing the attention for the distillation of this layer.

C. Training Paradigm

In order to learn excellent the hallucination stream, taking the missing SAR modality as an example, we consider the training paradigm of our proposed method from the perspective of curriculum learning in Fig. 1. Our method is divided

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into two main parts: the first part achieves the training of the teacher network in Fig. 1(b) and (c); the second part achieves the training of the student network in Fig. 1(d) and (e).

1) Training Step 1: In Fig. 1(b), the optical and SAR streams are training separately based on the cross-entropy loss function as follows:

$$L_{\text{opt}} = -\frac{1}{N}\sum_{n=1}^{N} l_n \log(\tilde{p}_n) \tag{8}$$

$$L_{\text{sar}} = -\frac{1}{N}\sum_{n=1}^{N} l_n \log(\hat{p}_n) \tag{9}$$

where $\tilde{p}_n$ is the softmax probability of the nth sample n in the optical stream, $\hat{p}_n$ is the softmax probability of the nth sample in the SAR stream, l_n denotes to the true label of the nth sample, and N is the number of samples in the mini-batch.

2) Training Step 2: In Fig. 1(c), the second training step is still further to train the teacher model, which initializes optical and SAR streams with weights learned in step 1. The joint cross-entropy is used as loss function.

$$L_{\text{opt+sar}} = -\frac{1}{N}\sum_{n=1}^{N} l_n \log\left(\tilde{p}'_n \oplus \hat{p}'_n\right) \tag{10}$$

where $\tilde{p}'_n$ is the softmax probability of the nth sample in the for optical stream in Fig. 1(c), $\hat{p}'_n$ is the softmax probability of the nth sample for the SAR stream in Fig. 1(c).

3) Training Step 3: We pay attention to the training of the hallucination stream, where the hallucination stream is initialized with the weights of optical stream in step 2, and the SAR stream is initialized with the weights of SAR stream in step 2 and then are frozen. The network is trained with the joint loss function as follows:

$$L_{\text{hall+sar}} = \alpha L_{kd} + \beta L_{\text{hall}} + \left(1 - \frac{\alpha+\beta}{2}\right) L_{\text{dis}} \tag{11}$$

where $L_{kd} = -(1/N)\sum_{n=1}^{N} O_n^{S\tau} \log(O_n^{T\tau})$ is the knowledge distillation (KD) loss, which has been proposed by Hinton *et al.* [42] to soften these probabilities using temperature scaling τ, $O_n^{T\tau} = \text{softmax}((\tilde{p}''_n/\tau))$, $O_n^{S\tau} = \text{softmax}((\hat{p}''_n/\tau))$, and $\hat{p}''_n$ is the softmax probability of the $n$thsample for the SAR stream in Fig. 1(c). $L_{\text{hall}} = -(1/N)\sum_{n=1}^{N} l_n \log(\hat{p}''_n)$, $\hat{p}''_n$ is the softmax probability of the nth sample for the hallucination stream in Fig. 1(c), α and β are the balance coefficients.

4) Training Step 4: In Fig. 1(e), we initialize the hallucination stream with the weights of hallucination stream from step 2, and initialize optical stream with the weights of optical stream from step 2. In the step, only the optical samples are uses as the inputs. We fine-tune the joint model composed by the hallucination and optical streams using joint cross-entropy as follows:

$$L_{\text{opt+hall}} = -\frac{1}{N}\sum_{n=1}^{N} l_n \log\left(\tilde{p}'''_n \oplus \hat{p}'''_n\right) \tag{12}$$

where $\tilde{p}'''_n$ is the softmax probability of the nth sample for the optical stream in Fig. 1(e), $\hat{p}'''_n$ is the softmax probability of the nth sample for the hallucination stream in Fig. 1(e). In a word, the training paradigm of the proposed DH-ADNet with missing SAR modality is shown in Algorithm 2.

Algorithm 2 The Training Paradigm of the proposed DH-ADNet (Taking Completely Missing SAR Modality as an Example)

Training Step 1 [in Fig. 1(b)]:
1) initialize optical and SAR streams with pretrained weights;
2) train optical and SAR streams based on (8) and (9);
Training Step 2 [in Fig. 1(c)]:
1):initialize optical and SAR streams with weights learned in step 1;
2): train both streams based on (10).
Training Step 3 [in Fig. 1(d)]:
1): initialize optical stream with optical weights learned in step 2 and then are frozen;
2): initialize hallucination stream with SAR weights learned in step 2;
3): train the composition i based on (11);
Training Step 4 [in Fig. 1(e)]:
1): initialize the hallucination stream with weights learned in step 3;
2): initialize optical stream with optical weights from step 2;
3): fine-tune firth composition based on (12).

TABLE II
LAND TYPES DISTRIBUTION ON THE PODELTA AND LIBOURNE

Land Types	PoDelta		Libourne	
	Number	Percentage	Number	Percentage
Resident area	213868	2.99%	3999047	17.97%
Water (Lake)	826053	11.55%	355261	1.60%
Background (Road)	497418	6.95%	723753	3.25%
Farmland	3834510	53.58%	11890375	53.41%
Tree	1784183	24.93%	2949197	13.25%
Industrial land	–	–	914262	4.11%
River	–	–	1426033	6.41%

IV. EXPERIMENTS AND DISCUSSION

A. Data Sets

The PoDelta is the first multimodal land cover classification data set, which is from the region of PoDelta. The PoDelta has been used to land cover classification in the literature [4] and [33], which has been confirmed the credibility. The PoDelta contains five land cover categories: resident area, farmland, water, tree, and background, as indicated in Table II. The size is 3666×1952, Fig. 5(b) shows the SAR image with HH-polar imaging mode, the corresponding optical images from GoogleEarth are shown in Fig. 5(a).

The Libourne is the second multimodal land cover classification data set, which is from the region of Libourne city, Bordeaux, France. The sizes of Libourne are 6384×3542, and the resolution is 3.0 m $\times$ 3.0 m. In which, Fig. 6(b) shows the SAR images with HH-polar imaging mode from the TerraSAR is geometrically rectified multispectral Level-1C reflectance products. The corresponding optical image from the GoogleEarth is shown in Fig. 6(a). In order to evaluate the proposed DH-ADNet, we can obtain the ground truth maps of Libourne based on the Open Street Maps [46] and

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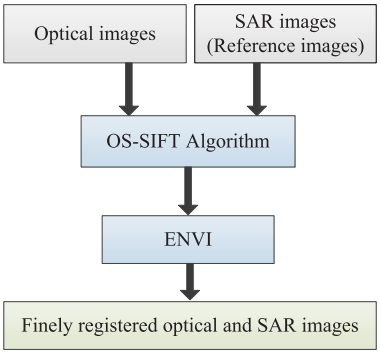


Fig. 3. Coregistration flowchart of the optical and SAR images.

the visual interpretation, as shown in Fig. 6(c). The contains Libourne seven land cover categories: resident area, river, road, farmland, tree, industrial area, and lake.

For the coregistration of optical and SAR images on the above two data sets, we adopt the coarse-to-fine registration strategy to complete fine registration. Fig. 3 shows the coregistration flowchart of the optical and SAR images. First, we download optical images of level 17 from Google Earth, whose spatial resolution 2.15 m × 2.15 m are slightly lower than SAR image. Then, the OS-SIFT algorithm [52] is used for the coarse registration, when taking the SAR image as the reference image. Finally, ENVI software achieves fine registration based on lots of corresponding points between the optical and SAR images are selected manually.

B. Implementation Details

All experiments and analyses are conducted by Pytorch library on the platform of Window 10 with Intel(R) Corn(TM) i7-8700K 3.70-GHz CPU, NVIDIA GeForce GTX 2080Ti GPU and 32-GB memory. The entire framework is trained by using Adam optimizer with the batch size 256. In Fig. 1(b), the epoch is 80, and the initial learning rate is 1.0×10^{-4}, In Fig. 1(c), the epoch is 80, and the initial learning rate is 1.0×10^{-5}. In Fig. 1(d), the epoch is 50, and the initial learning rate is 1.0×10^{-4}. And the epoch is 80, and the initial learning rate is 1.0×10^{-5} in Fig. 1(e).

C. Comparison Methods

In this section, we apply three categories of methods, in which the backbone is AlexNet without a fully connected layer.

1) Variant Models: To illustrate our proposed DH-ADNet can efficiently realize land cover classification within missing modalities, we compare the DH-ADNet with the following variant models.

1) CNN-Opt.: The optical stream network in Fig. 1(b) is used as a reference, only optical modalities are used as the inputs of CNN-Opt. for land cover classification.
2) CNN-SAR: The SAR stream network in Fig. 1(b) is used as a reference,only SAR modalities are used as the inputs of CNN-SAR for land cover classification.
3) ADNet: Zagoruyko and Komodakis [31] have verified the effectiveness of attention mechanism in the distillation network. Therefore, to adapt such a model to our scenario, attention distillation proposed by Zagoruyko and Komodakis [31] is used to replace the DH-ADM in the DH-ADNet, and which is termed as the ADNet.

2) Multimodal Deep Models: We list the performance of the multimodal deep learning models that can fully explore both optical and SAR modalities within complete modalities.

1) CNN-Opt.&SAR: Multimodal network in Fig. 1(c) is used as a reference, which is trained based on the optical and SAR modalities for land cover classification.
2) 2-D-CNN: Kussul *et al*. [47] proposed a novel deep fusion methods to integrate both optical and corresponding SAR images. In this article, we apply the pretrained AlexNet as backbone for 2-D-CNN. And we use a randomly initialized classification layer with softmaxt function to replace the FC layer of AlexNet.
3) MBFNet: Li *et al.* [4] proposed a novel multimodal bilinear fusion network with second-order attention-based channel selection module (SACSM) to integrate both optical and SAR features. Where the SACSM reconfigured compact feature maps with better discrimination, the fusion features between optical and SAR features are obtained based on the bilinear pooling model.

3) State-of-the-Art Privileged Information Models: The three categories report the performances of other privileged information methods [7], [22], [48] within missing heterogeneity

1) Hall-Net: Hallucination network has been proposed to deal with RGB-D object detection when depth modality is missing [22]. After that, Kampffmeyer *et al*. [7] applied hallucination network for urban land cover classification with missing data modalities. To achieve comparison, we apply it to deal with the land cover classification with optical and SAR modalities.
2) Garcia *et al*. [48] proposed a novel approaches for the RGB-D vision tasks in the unified frameworks of distillation and privileged information. To achieve comparison, we use the method proposed by Garcia *et al*. [48] to deal with optical and SAR modalities.

D. Configuration of Hyperparameters

The size of training samples p_s, the number of training samples N_s, the number of centers M, the balance parameter λ in (3), and the balance parameters α and β in (11) influence the land cover classification performance of our proposed method. To explore the influence of the above parameters, the specific analysis of them on the PoDelta data set will be elaborated. The determining processes of parameters of the Libourne data set are the same as the experiment of the PoDelta.

(1) Influence of the patch size is p_s. In Fig. 4(a) and (b), we vary the p_s from 13×13 to 33×33 with a span of 4 and keep the other parameters fixed. It is seen that the p_s has an important influence on the land cover classification performance. When the p_s is increased, the curve of classification accuracies shows an increasing trend and then decreases, which is mainly because a too small p_s cannot contain richer spatial information and a too larger p_s contain

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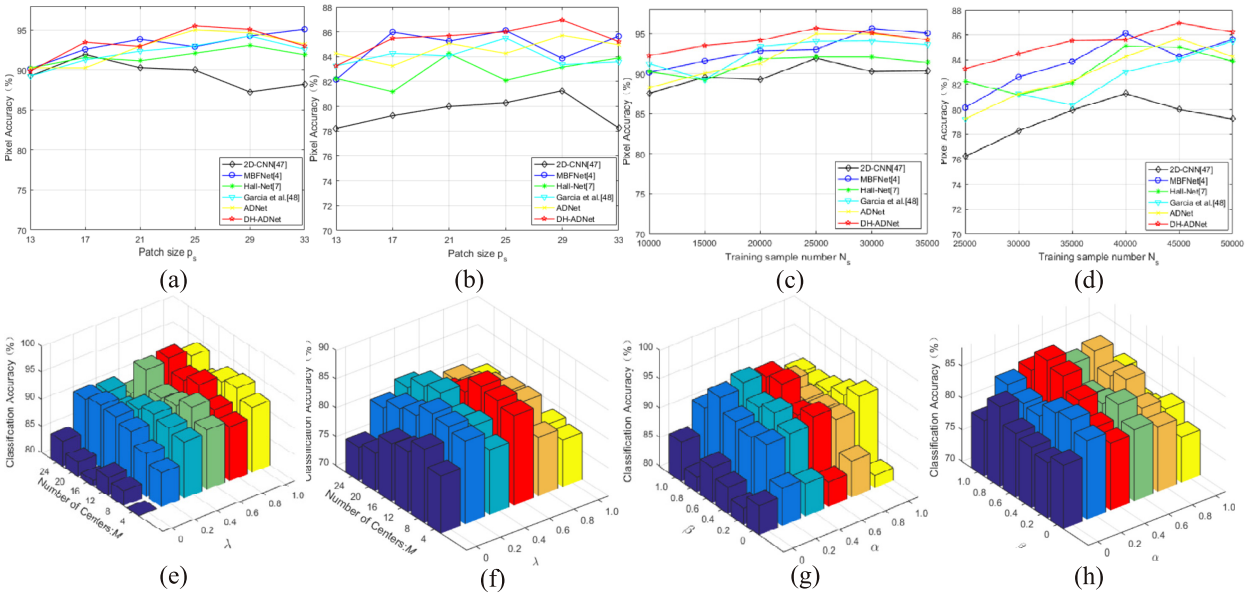


Fig. 4. Depiction the trend between the accuracy and the parameters patch size P_s, the number of training sample N_s, the number of centers M, λ, α, and β on PoDelta and Libourne data sets. (a) and (b) denote the accuracy curves with different patch size P_s on PoDelta and Libourne, respectively. (c) and (b) denote the accuracy curves with different number of training sample N_s on PoDelta and Libourne, respectively. (e) and (f) denote the accuracy curves with a different number of centers M and λ on PoDelta and Libourne, respectively. (g) and (h) denote the accuracy curves with α and β on PoDelta and Libourne, respectively.

TABLE III

LAND COVER CLASSIFICATION PRECISION FOR THE METHODS WITH AND WITHOUT MSIS ON THE PODELTA

Methods	MSIS	Test Mods.	PA	MPA	MIoU	FWIoU
CNN-SAR	✗	SAR	80.87±3.01	81.97±2.89	61.73±2.51	69.26±3.23
CNN-SAR	✓	SAR	**82.08±2.28**	**85.69±2.14**	**63.59±1.70**	**71.40±2.26**
CNN-Opt.	✗	Opt.	91.06±2.12	87.71±1.98	81.29±1.87	84.83±2.03
CNN-Opt.	✓	Opt.	**92.21±1.08**	**89.07±1.21**	**82.24±0.97**	**86.61±1.04**
CNN-Opt.&SAR	✗	Opt.+SAR	94.45±1.56	94.35±1.09	84.80±1.32	91.51±1.56
CNN-Opt.&SAR	✓	Opt.+SAR	**95.36±0.52**	**95.58±0.48**	**85.95±0.73**	**92.09±0.71**
DH-ADNet	✗	Opt.	94.34±1.23	95.02±0.89	82.98±1.34	91.09±1.23
DH-ADNet	✓	Opt.	**95.22±0.78**	**95.48±0.56**	**83.47±0.93**	**91.56±0.80**

the heterogeneous spatial information. From Fig. 4(a) and (b), we can see that setting $p_s = 25$ and $p_s = 29$ for the PoDelta and Libourne data sets is a good choice, which contain the homogeneous spatial information to generate more robust features.

(2) Influence of the training sample number N_s of each land cover categories. The amount of land cover categories is unbalanced on two broad coregistered optical and SAR data sets, and it is reasonable to adopt the selecting strategy that the identical samples amount of each land category is chosen in our experiments, which can ensure the unbiased training of the network. We vary N_s from 10 000 to 35 000 with a step of 5000 and keep the other parameters fixed in Fig. 4(c). And in Fig. 4(d), we vary the N_s from 25 000 to 50 000 with a step of 5000 and keep the other parameters fixed. From Fig. 4(c) and (d), we can find that the proposed DH-ADNet are not sensitive for N_s as the methods without the proposed MSIS, which confirm the MSIS can realize the selection of most representative multimodal instances. It can achieve a stable good enough results when $N_s = 25\,000$ and $N_s = 40\,000$ for the PoDelta and Libourne data sets.

(3) Influence of the parameters M and λ. The parameters M and λ of PoDelta and Libourne data sets are determined as shown in Fig. 4(e) and (f), where the horizontal axis denotes the parameter M, and its value ranges from 4 to 24 with step 4; The vertical axis denotes the parameter λ, and its value ranges from 0 to 1.0 with a step of 0.2. We see that the highest accuracy is achievable when the parameter values are $M = 20$ and $\lambda = 0.8$ in Fig. 4(e). Meanwhile, in Fig. 4(f), the classification precision is the optimal, when the parameter values are $M = 16$ and $\lambda = 0.6$.

(4) Influence of the parameters α and β. In (11), the parameters α and β provide an effective tradeoff between the KD loss L_{kd}, L_{dis}, and L_{hall}. In Fig. 4(g) and (h), we vary the α from 0 to 1.0 with a span of 0.2, and the value of the parameters β ranges from 0 to 1.0 with a span of 0.2. From Fig. 4(g), we can see that setting $\alpha = 0.4$ and $\beta = 0.6$ is a good choice for the PoDelta dataset, and we suggest to set $\alpha = 0.4$ and $\beta = 0.6$ as a compromise choice for the Libourne data set. In (11), the parameters α and β provide an effective tradeoff between the KD loss L_{kd}, L_{dis}, and L_{hall}. When $\alpha = 0$, $\beta = 0$ and $\alpha = \beta = 1$, the L_{kd}, L_{dis}, and L_{hall} in (11) are

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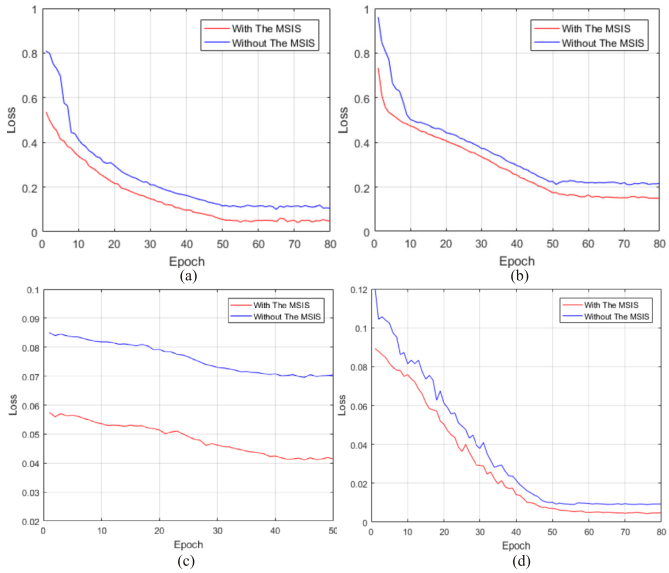


Fig. 5. Training loss curves of different methods with and without patch ranking. (a) CNN-Opt. (b)CNN-SAR. (c) CNN-Opt.&SAR. (d)DH-ADNet.

TABLE IV
LAND COVER CLASSIFICATION PRECISION WHEN THE DH-ADM IS EMBEDDED INTO DIFFERENT LAYERS OF THE DH-ADNET ON THE PODELTA

#	Methods	Test Mods.	*LRN_1*	*LRN_2*	*Conv3_3*	AP
1	CNN-Opt.	Opt.				92.07±0.90
2	DH-ADNet	Opt.	✓			91.68±0.89
3	DH-ADNet	Opt.		✓		93.05±0.86
4	DH-ADNet	Opt.			✓	93.53±0.82
5	DH-ADNet	Opt.	✓	✓		92.94±0.80
6	DH-ADNet	Opt.	✓		✓	94.21±0.85
7	DH-ADNet	Opt.		✓	✓	94.78±0.76
8	DH-ADNet	Opt.	✓	✓	✓	**95.22±0.78**

ignored, respectively, whose precisions are below that of the same methods under $\alpha \neq 0$, $\beta \neq 0$, and $\alpha \neq \beta \neq 1$, which means each component plays a certain role in the land cover classification.

E. Ablation Experiments

It is worth to evaluate the effectiveness of the DH-ADM and MSIS in our model. We conduct extensive ablation experiments on the PoDelta data set with different settings for the DH-ADNet.

1) Effect of the MSIS: To explore the effectiveness of the proposed MSIS, we analyze the convergence of training loss and the accuracy of classification of the CNN-SAR, CNN-Opt., CNN-Opt.&SAR and DH-ADNet. From Section IV-D, we know that the N_s and p_s are 25 000 and 25×25, respectively. Table III lists the comparison results between the above methods with and without the MSIS. One can see that the methods with MSIS perform better land cover classification than the same method without the MSIS on the PoDelta data set, which adequately demonstrates the effectiveness of the proposed MSIS strategy and its positive effect on improving the discrimination of the proposed DH-ADNet.

Furthermore, we compare the two training loss curves with and without the MSIS of the CNN-SAR, CNN-Opt., CNN-Opt.&SAR, and DH-ADNet in Fig. 5. We clearly find that the loss curves of the methods with the MSIS are consistently lower than those of other methods without the MSIS, which indicates that the MSIS can select the representative optical and SAR samples to accelerate the convergence of the proposed DH-ADNet. Therefore, a more discriminative feature can be learned by the DH-ADNet. In summary, it is clear that the proposed MSIS realizes the selection of most representative multimodal instances instant of the dirty samples for enhancing the discriminative power of the feature space extracted by the DH-ADNet.

2) Effect of the DH-ADM: The proposed DH-ADM is used to distill the feature maps from *LRN_1*, *LRN_2*, and *Conv3_3* layer of hallucination and SAR streams in Fig. 1(d). To analyze whether the DH-ADM is able to automatically highlight different contributions of multilayer distillation by carefully exploring the classification losses of different layer from hallucination over the training iterations, we explore to place the DH-ADM after different layer in Table IV. Where we find that overall adding the hallucination loss at mid-to-lower layer improved performance than the optical modality only baseline network, CNN-Opt., except for the method in Row #2, because there is a larger heterogeneous gap between low-layer features of optical and SAR modalities. And some phenomena can be observed in Table IV.

1) Adding the DH-ADM at different layers can obtain good land cover classification, which indicates that different layer distillation has some contributions.
2) Adding the DH-ADM at a deeper layer can obtain better land cover classification (shown in rows #2, #3, and #4), which indicates that the deeper feature can better reflect essential information of the data.
3) Adding the DH-ADM at the multilayer can obtain better land cover classification than adding the DH-ADM at single layer, which indicates that the DH-ADM can automatically learn the varying contributions for multilayer distillation. The highest overall performance was achieved with the DH-ADM on the *LRN_1*, *LRN_2*, and *Conv3_3* layer.

F. Experimental Results

To verify the effectiveness of the DH-ADNet, the mean pixel accuracy (MPA), the mean intersection over union (MIoU) and the frequency weighted intersection over union (FWIoU) are used to evaluate the classification performance, and in order to keep the land cover classification results stable, the experiments are repeated ten times under random training samples. Figs. 6 and 7 show the classification maps of different methods on the PoDelta and Libourne data sets. Furthermore, confusion matrixes are present in Fig. 8.

1) Classification Results on PoDelta: We first presented the land of cover classification different methods in Table V, when the SAR images are missing. Meanwhile, the results without optical modality presented in Table VI, the hallucination stream distills knowledge from the optical stream in Fig. 1(d).

a) SAR Modality Missing: From Table V, we have gathered several experiment findings. First, rows #1 and #2 present the performance of the separate optical and SAR streams

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TABLE V

PA, MPA, MIoU, AND FWIoU OF DIFFERENT METHODS ON PADELTA WHEN THE SAR IMAGES ARE MISSING. 2D-CNN*, MBFNET*, HALL-NET* AND GARCIA *et al.** DENOTE 2D-CNN, MBFNET, HALL-NET AND GARCIA *et al.* WITH MSIS, INSTEAD OF SAMPLE RANDOM SELECTION STRATEGY

#	Methods	Test Mods.	PA	MPA	MIOU	FWIoU
1	CNN-SAR	SAR	**82.08±2.28**	**85.69±2.14**	**63.59±1.70**	**71.41±2.26**
2	CNN-Opt.	Opt.	**92.21±1.08**	**92.07±1.21**	**82.24±0.97**	**86.62±1.04**
3	CNN-Opt.&SAR	Opt.+SAR	**95.36±0.52**	**95.58±0.48**	**85.95±0.73**	**92.09±0.71**
4	2D-CNN[47]	Opt.+SAR	91.94±1.23	88.42±1.32	82.16±1.14	85.99±0.99
5	2D-CNN*	Opt.+SAR	92.54±0.98	89.21±1.12	82.99±1.00	87.01±0.82
6	MBFNet [4]	Opt.+SAR	95.09±1.26	95.04±1.09	83.22±1.47	91.49±1.01
7	MBFNet*	Opt.+SAR	96.18±0.82	96.35±0.74	84.21±0.80	92.03±0.75
8	Hall-Net [7]	Opt.	93.09±1.31	93.19±0.82	81.38±1.51	89.42±0.92
9	Hall-Net*	Opt.	93.58±1.02	93.79±0.53	81.75±0.72	90.02±0.48
10	Garcia et al.[48]	Opt.	94.35±1.21	94.33±1.03	82.43±1.42	90.77±1.09
11	Garcia et al.*	Opt.	94.85±0.93	94.74±0.62	82.99±1.01	91.07±0.64
12	ADNet	Opt.	94.61±0.82	94.46±0.52	82.52±0.76	90.98±0.85
13	DH-ADNet	Opt.	**95.22±0.78**	**95.48±0.56**	**83.47±0.93**	**91.56±0.80**

TABLE VI

PA, MPA, MIoU, AND FWIoU OF DIFFERENT METHODS ON PADELTA WHEN THE OPTICAL IMAGES ARE MISSING

#	Methods	Test Mods.	PA	MPA	MIOU	FWIoU
1	CNN-SAR	SAR	**82.08±2.28**	**85.69±2.14**	**63.59±1.70**	**71.41±2.26**
2	Hall-Net [7]	SAR	83.71±1.51	86.94±1.65	66.45±1.52	73.53±1.68
3	Hall-Net*	SAR	84.31±1.02	87.54±1.25	67.85±1.32	74.29±1.01
4	Garcia et al.[48]	SAR	85.21±1.42	88.75±1.52	68.05±1.67	75.89±1.89
5	Garcia et al.*	SAR	85.96±0.99	88.84±1.09	71.55±1.21	76.77±1.45
6	ADNet	SAR	85.72±1.03	88.56±0.98	71.13±1.34	76.35±1.12
7	DH-ADNet	SAR	**86.99±0.97**	**89.79±1.11**	**72.04±1.01**	**77.98±0.99**

in Fig. 1(b), of course, we expect our proposed model (shown in row #13) and obtain better perform than the one trained on optical modality only, with 3.01% improvement in the PA, meanwhile, our proposed model obtain lower standard deviation, which verifies the features that are extracted the proposed DH-ADNet have more discriminative and can reduce the uncertainty of classification. And we also propose our reference, the CNN-Opt.&SAR, consisting in the teacher model trained in Fig. 1(c), the proposed DH-ADNet trained on PoDelta achieves the slightly lower performance and standard deviation compared to the CNN-Opt.&SAR, which verifies the proposed DH-ADNet is able to imitate the significant information contained in SAR modality that is missing at test time for land cover classification. Second, the 2D-CNN [47] and MBFNet [4] are multimodal deep learning models without the MSIS, which randomly selected the training samples as inputs. Compared to our models performance, the PAs of 2D-CNN and MBFNet are reduced by 3.28% and 0.13%, respectively. And when the training samples extracted by the MSIS are used as inputs to MBFNet*(in row #7) that has 96.18% PA, increased by 1.08% compared to original MBFNet, which verifies the MSIS can select most representative instances. Third, rows #9 and #11 reports the other privileged information methods without the DH-ADM [7], [48], Hall-Net* and Garcia *et al.**, in order to keep fair direct comparison, we can make same training and test data, we can see that our privileged information approach outperforms. Meanwhile, ADNet is the mutation of our model without the DH-ADM, instant attention distillation proposed by [31], the proposed DH-ADNet achieves a better land cover classification performance than Hall-Net* and Garcia *et al.** and ADNet, which verifies the proposed DH-ADM not only is able to perform multilayer distillation between the SAR and hallucination streams but also can adaptive highlighting different contributions of multilayer distillation.

Classification maps of different methods are shown in Fig. 6. We can find that the classification map of our method is better than other contrast methods, except for CNN-Opt.& SAR. In Fig. 6(d)–(e), the classification maps based on single-modal data are the worst. Fig. 6(f)–(l) presents the classification maps based on multimodal data, which are better than the classification maps shown in Fig. 6(d) and (f). The optical and SAR have complementary information that is good for land cover classification, and Fig. 6(i)–(l) can deal with land cover classification with missing modality. Especially, Compared with several privileged methods, the land cover classification map obtained by our method is better.

b) Optical Modality Missing: When the optical modality is missing, the hallucination stream distills knowledge from the optical stream in Fig. 1(d). Four quantitative metrics of our model and comparison methods are presented to objectively evaluate the performance of our method in Table VI. It is discovered that the DH-ADNet has much better accuracies than the comparison methods.

In particular, we observe that the performance of the proposed DH-ADNet degrades by almost 5%, 86.99% versus

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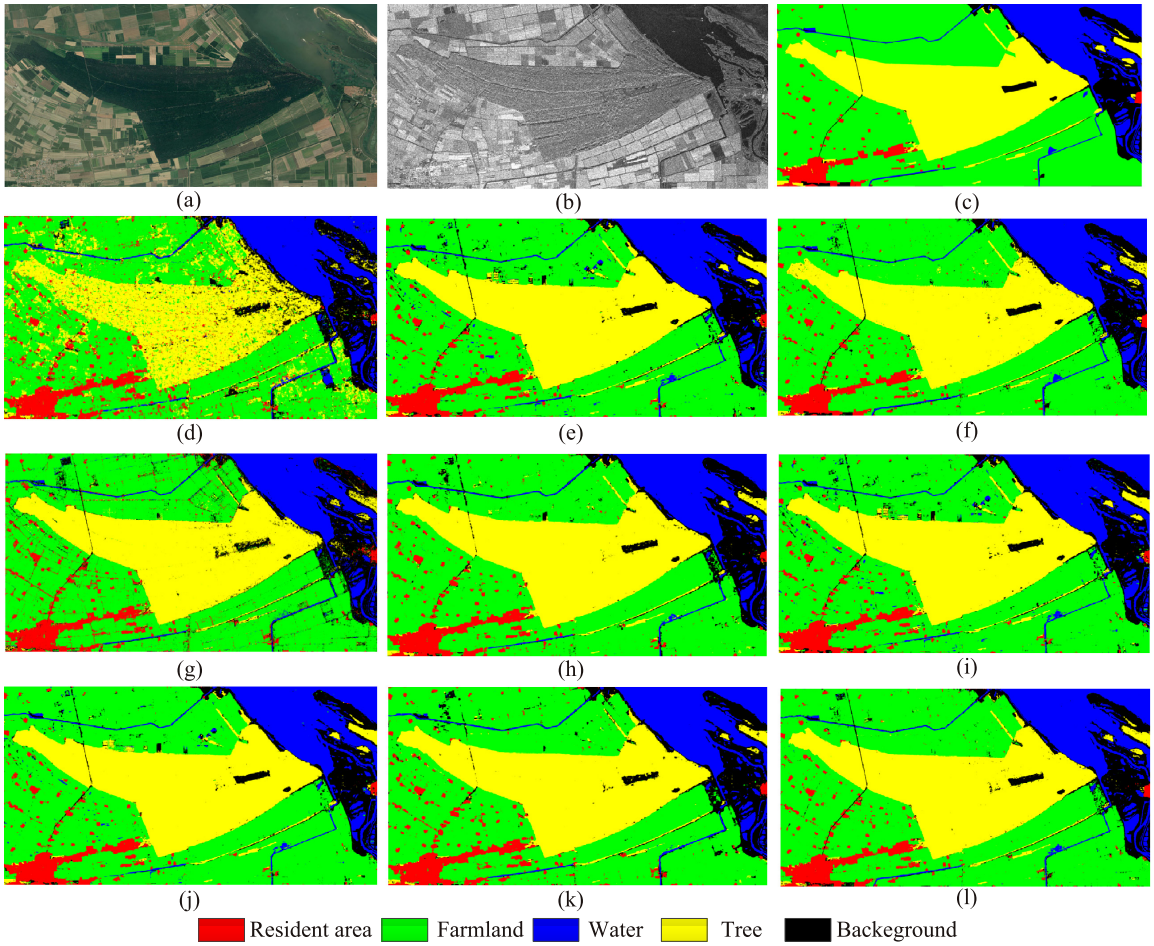


Fig. 6. Classification maps of different methods on PoDelta. (a) Original optical image. (b) Original SAR image. (c) Ground truth. (d) CNN-SAR. (e) CNN-Opt. (f) CNN-Opt.& SAR. (g) 2D-CNN[9]. (h) MBFNet *et al.* [4]. (i) Hall-Net [7]. (j) Garcia *et al.* [48]. (k) ADNet. (l) DH-ADNet.

82.08% (CNN-SAR of Table VI), which indicates DH-ADNet can learning the privilege information. Meanwhile, we compare our method with the privileged information methods, Hall-Net and Garcia *et al.*, and variant models, ADNet, the DH-ADNet achieves better land cover classification performance.

2) Classification Results on Libourne: First, we presented the land cover classification of different methods in Table VII when the SAR images are missing. Meanwhile, the classification results are presented in Table VIII when the optical modalities are missing, the hallucination stream distills knowledge from the optical stream in Fig. 1(d).

a) SAR Modality Missing: Table VII lists the PA, MPA, MIoU, and FWIoU of land cover classification. First, it is obvious that the proposed DH-ADNet achieves a better land cover classification performance compared with unimodal deep learning models, for instance CNN-SAR and CNN-Opt., with 18.86% improvement, which verifies the DH-ADNet can achieve preferable extraction of the privileged information and then obtain better land cover classification during modal missing. Second, rows #3 to rows #7 list the performance of the multimodal land cover classification algorithms, we can find that the proposed DH-ADNet obtains a slightly lower performance than the CNN-Opt.&SAR, which verifies the DH-ADNet is able to imitate the significant information contained in SAR modality that is missing at test time for land cover classification. Rows #4 and #5 denote two 2D-CNN [47] with and without MSIS, it is obvious that the performance of 2D-CNN with MSIS is better than the 2D-CNN with random sample selection, therefore, the MSIS is used to select a representative sample that is important. Similarly, in rows #6 and #7, we also find the performance of MBFNet [4] with MSIS is better the MBFNet without MSIS. Moreover, by compared rows #6, #4, and #13, we can find that the proposed DH-ADNet is better than multimodal deep land cover classification methods. Third, Hall-Net [31] and Garcia *et al.* [7] as the novel distillation models, it is seen that our method can obtain better land cover classification results, mainly because the proposed DH-ADM embedded into our method can dynamically realize hierarchical attention distillation. Finally, in ADNet, the weights of distillation loss for *LRN_1*, *LRN_2* and *Conv3_3* are set to equal, whose results are worse than our method. Meanwhile, by comparing with a standard deviation of different methods, the proposed DH-ADNet have a slightly higher standard deviation compared to the CNN-Opt.&SAR and MBFNet*, which verifies the proposed DH-ADNet is able to imitate the significant information contained in missing modality that is missing at

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TABLE VII
PA, MPA, MIOU AND FWIOU OF DIFFERENT METHODS ON LIBOURNE WHEN THE SAR IMAGES ARE MISSING

#	Methods	Test Mods.	PA	MPA	MIOU	FWIoU
1	CNN-SAR	SAR	**68.09±3.08**	**72.08±2.54**	**49.48±2.73**	**53.79±2.52**
2	CNN-Opt.	Opt.	**80.75±2.08**	**86.08±1.95**	**67.09±2.28**	**68.01±1.99**
3	CNN-Opt.&SAR	Opt.+SAR	**87.21±1.42**	**91.37±1.38**	**75.47±1.70**	**77.96±1.46**
4	2D-CNN[47]	Opt.+SAR	80.15±1.51	85.09±1.69	66.15±2.01	67.71±2.12
5	2D-CNN*	Opt.+SAR	81.25±1.28	86.59±1.48	68.05±1.85	68.51±1.56
6	MBFNet [4]	Opt.+SAR	86.12±1.75	90.98±1.59	74.28±2.15	76.95±2.01
7	MBFNet *	Opt.+SAR	88.31±1.33	92.57±1.36	76.48±1.42	78.75±1.34
8	Hall-Net [7]	Opt.	84.25±1.69	87.57±1.87	71.56±1.93	75.08±1.85
9	Hall-Net *	Opt.	85.05±1.51	88.26±1.62	72.36±1.59	75.69±1.44
10	Garcia et al.[48]	Opt.	85.50±1.71	88.94±1.77	73.34±2.01	76.04±1.90
11	Garcia et al.*	Opt.	85.98±1.57	89.68±1.60	74.03±1.84	76.70±1.66
12	ADNet	Opt.	85.71±1.65	89.05±1.60	73.57±1.85	76.21±1.78
13	DH-ADNet	Opt.	**86.95±1.52**	**90.34±1.56**	**75.08±1.76**	**77.92±1.58**

TABLE VIII
PA, MPA, MIOU AND FWIOU OF DIFFERENT METHODS ON LIBOURNE WHEN THE OPTICAL IMAGES ARE MISSING

#	Methods	Test Mods.	PA	MPA	MIOU	FWIoU
1	CNN-SAR	SAR	**68.09±3.08**	**72.08±2.54**	**49.48±2.73**	**53.79±2.52**
2	Hall-Net [7]	SAR	71.14±2.55	73.05±2.04	57.25±2.58	56.38±2.31
3	Hall-Net*	SAR	72.24±2.04	74.57±1.85	57.96±1.77	57.56±1.98
4	Garcia et al.[48]	SAR	72.56±2.79	74.98±2.54	58.75±2.61	58.12±2.31
5	Garcia et al.*	SAR	73.65±2.19	75.58±2.08	59.87±1.99	60.48±1.81
6	ADNet	SAR	73.58±2.23	75.24±1.89	59.59±2.32	60.42±1.78
7	DH-ADNet	SAR	**74.12±2.08**	**76.25±1.88**	**60.58±2.28**	**61.49±1.69**

test time, and obtain robust fusion features for land cover classification.

Fig. 6 shows the land cover classification maps of different methods. The classification map derived from the proposed DH-ADNet is better than some contrast methods, especially CNN-SAR, CNN-Opt., and 2D-CNN. Taking the road as an example, it can be seen that the road derived from our DH-ADM method is more complete and has fewer misclassified.

b) Optical Modality Missing: When the optical modality is missing, the hallucination stream distills knowledge from the optical stream in Fig. 1(d). In Table VIII, four quantitative metrics of our model and comparison methods are presented to assess the land cover classification performance of the proposed DH-ADNet. In particular, the performance of the proposed DH-ADNet degrades by almost 6%, 74.12% versus 68.09% (CNN-SAR in Table VIII), which indicates the DH-ADNet can provide learning privilege information. Meanwhile, by comparing privileged information methods, Hall-Net and Garcia *et al*., and variant models, ADNet, the DH-ADNet can achieve a better performance.

3) Computational Complexity: The proposed DH-ADNet all consist of four compositions shown in Fig. 1(b)–(e). The Hall-Net* [7], Garcia *et al.** [48], ADNet as the variants of the proposed DH-ADNet, which also consist of four compositions. The parameter number and FLOPs of each composition from the Hall-Net* [7], Garcia *et al.** [48], ADNet, and the proposed DH-ADNet shown in Table IX on the PoDelta and Libourne data sets, respectively. In Table IX, the "Comput. Comp." denotes the

TABLE IX
PARAMETERS NUMBER AND FLOPS OF DIFFERENT METHODS ON PADELTA AND LIBOURNE DATA SETS

Datasets	Methods	Comput. Comp.	Compositions			
			S1	S2	S3	S4
PoDelta	Hall-Net*	Para.(M)	18.3	18.2	18.2	18.2
		FLOPs(M)	472.7	462.2	462.2	462.2
	Garcia et al.*	Para.(M)	18.3	18.2	18.2	18.2
		FLOPs(M)	472.7	462.2	462.2	462.2
	ADNet	Para.(M)	18.3	18.2	19.3	18.2
		FLOPs(M)	358.6	350.5	350.5	350.5
	DH-ADNet	Para.(M)	18.3	18.2	19.6	18.2
		FLOPs(M)	358.6	350.5	374.7	350.5
Libourne	Hall-Net*	Para.(M)	18.5	18.4	18.4	18.4
		FLOPs(M)	270.9	259.6	259.6	259.6
	Garcia et al.*	Para.(M)	18.5	18.4	18.4	18.4
		FLOPs(M)	397.9	386.7	386.7	386.7
	ADNet	Para.(M)	18.5	18.4	19.5	18.4
		FLOPs(M)	365.1	353.8	353.8	353.8
	DH-ADNet	Para.(M)	18.5	18.4	19.8	18.4
		FLOPs(M)	365.1	353.8	383.4	353.8

"Computational Complexity" that includes the number of model parameters and FLOPs, and the "S1," "S2," "S3," and "S4" denote the four compositions in Fig. 1(b)–(e), respectively. First, the parameter number of "S1," "S2," and"S4" compositions from different methods on the PoDelta and Libourne data sets is the same, and the parameter number of "S3" composition from the proposed DH-ADNet has a slight increase due to the introduction of DH-ADM. Then, there is a certain correlation between FLOPs and image patch size, the proposed DH-ADNet and ADNet have the same patch size when they obtain optimal land cover classification, it is seen that the FLOPs of "S3" composition

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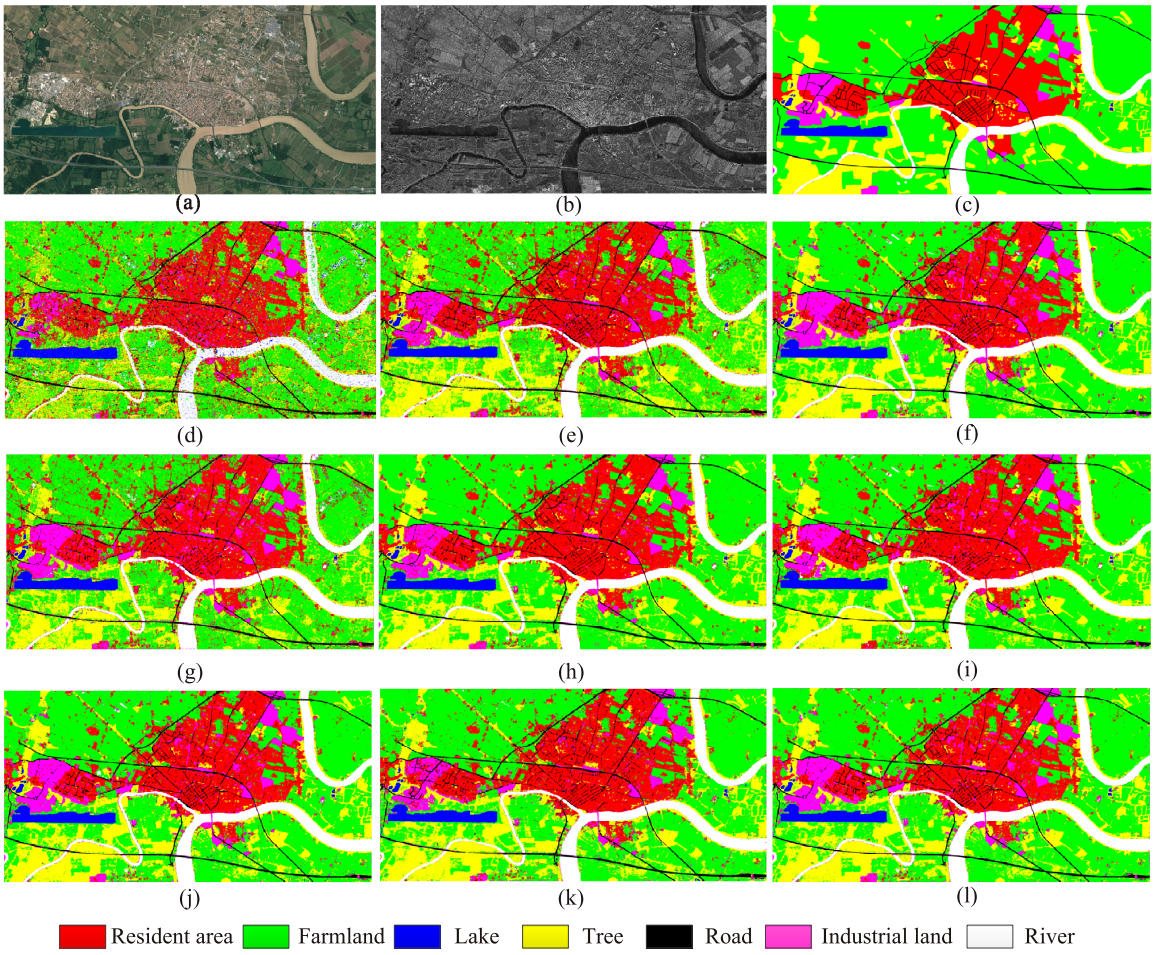


Fig. 7. Classification maps of different methods on Libourne. (a) Original optical image, (b) Original SAR image, (c) Ground truth, (d) CNN-SAR, (e) CNN-Opt., (f) CNN-Opt.& SAR, (g) 2D-CNN[9], (h) MBFNet *et al.* [4], (i) Hall-Net [7], (j) Garcia *et al.* [48], (k) ADNet, (l) DH-ADNet.

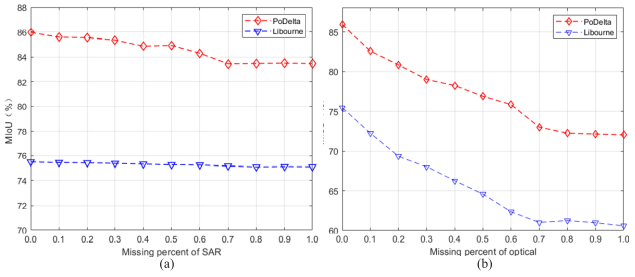


Fig. 8. Curves to depict the trend of classification accuracies with the missing percent of SAR or optical on PoDelta and Libourne data sets, respectively. (a) Accuracy curves is given based on different the missing percent of SAR modality on the PoDelta data set. (b) Accuracy curves are given based on different the missing percent of optical modality on the Libourne data set.

from the proposed DH-ADNet has a slight increase but instead of a larger increase because the DH-ADM adopts 1×1 convolution and global pooling. In a word, compared with the state-of-the-art privileged methods, the DH-ADM improves the performance of land cover classification with completely missing modalities, although it introduces an acceptable increase of modal parameters and the FLOPs.

4) Influence of Modality Missing Percent: To learn the influence of modality missing percent for land cover classification, we divide the test samples into two parts: test samples without missing modalities and test samples with completely missing modalities, the second composition of our proposed DH-ADNet [shown in Fig. 1(c)] complete the classification of test samples without missing modalities, and firth composition [shown in Fig. 1(e)] completes the land cover classification of test samples with completely missing modalities. The final land cover classification result is a combination of the results of the two compositions. The curve of the classification accuracies with the missing percent of SAR or optical is presented in Fig. 8 on the PoDelta and Libourne data sets, respectively.The horizontal axis denotes the missing percent of SAR and optical, and its value ranges from 0.0 to 1.0 with a span of 0.1, the vertical axis denotes the MIoU.

In Fig. 8(a), taking the red curve as an example, it is easy to find that the MIoU has reduced a little with missing percent from 0.0 to 0.7, and the classification accuracies are almost unchanged from 0.7 to 1.0. Our proposed methods can efficiently imitate the significant information contained in missing SAR modality, and it has robustness classification results with different missing percent of SAR modality. Similarly, we also can obtain a similar conclusion from other curves in Fig. 8(a). Furthermore, the optical modality is a

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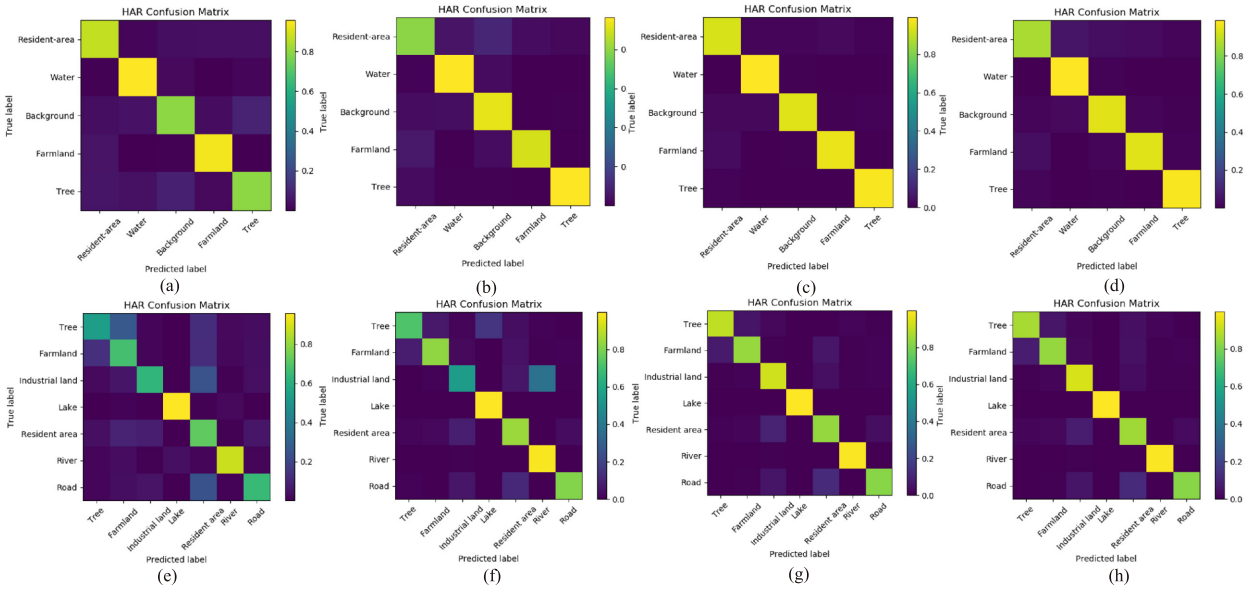


Fig. 9. Confusion matrixes of the proposed DH-ADNet on PoDelta and Libourne. (a), (b), (c), and (d) are the confusion matrixes of CNN-SAR, CNN-Opt. CNN-Opt. & SAR, and DH-ADNet on the PoDelta, respectively. (e), (f), (g), and (h) are the confusion matrixes of CNN-SAR, CNN-Opt. CNN-Opt. & SAR, and DH-ADNet on the Libourne, respectively.

dominant modality for land cover classification on the PoDelta and Libourne data sets, the missing optical modality influences the final classification to a certain extent. Meanwhile, it is a challenging problem to imitate optical features with SAR features due to poor interpretability of SAR images. Therefore, different from the curves in Fig. 8(a), the MIoU is declining with the missing percent of optical modality from 0.0 to 1.0 in Fig. 8(b). However, the proposed DH-ADNet also can imitate partly significant information of optical modality with SAR modality, and obtain better results than single-modal method (DH-ADNet versus CNN-SAR shown in Tables VI and VIII).

5) Confusion Matrixes: For further analysis, we illustrate the confusion matrixes of our method on the PoDelta and Libourne data sets in Fig. 9, which shows the confusion matrixes of CNN-SAR, CNN-Opt., CNN-Opt.&SAR, and the proposed DH-ADNet. From the confusion matrixes, the observations can be found: First, in Fig. 9(a) and (e), the most land cover types of PoDelta and Libourne data sets are misclassified by CNN-SAR. Then, the diagonal of confusion matrixes in Fig. 9(c) and (g) is highlighted, and CNN-Opt.&SAR obtain better land cover classification preformation. Finally, by compared Fig. 9(c), (d), (g), and (h), it is seen that confusion matrixes of CNN-Opt.&SAR and the proposed DH-ADNet is approximative, which proves our method can handle the land cover classification within modalities missing.

V. Conclusion

As two different tools of earth observation, optical and SAR data modalities can provide complementary information that led to more robust and better performance. However, not all modalities included in the training data sets could be available in testing samples. Therefore, when the modalities are missing at test time, it is important to consider how to learn robust representations leveraging multimodal data in the training stage. In this article, we propose a novel DH-ADNet with MSIS for land cover classification using missing data modalities. First, the MSIS realizes the selection of the most representative multimodal instances to enhance the DH-ADNet's ability of discriminative feature extraction. Then, the DH-ADNet is proposed to promote the hallucination stream learn the privileged information. Particularly, a novel DH-ADM is introduced to adaptively highlight different contributions of multilayer attention distillation by carefully exploring the classification loss of different layer features over the training iterations. Comprehensive evaluations on two coregistered optical and SAR data sets and report state-of-the-art results in the privileged information scenario.

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References

[1] J. Tian, S. Cui, and P. Reinartz, "Building change detection based on satellite stereo imagery and digital surface models," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 406–417, Jan. 2014.

[2] F. T. Ulaby, R. Y. Li, and K. S. Shanmugan, "Crop classification using airborne radar and landsat data," *IEEE Trans. Geosci. Remote Sens.*, vol. GE-20, no. 1, pp. 42–51, Jan. 1982.

[3] N. Kussul, G. Lemoine, F. J. Gallego, S. V. Skakun, M. Lavreniuk, and A. Y. Shelestov, "Parcel-based crop classification in ukraine using Landsat-8 data and sentinel-1A data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 6, pp. 2500–2508, Jun. 2016.

[4] X. Li, L. Lei, Y. Sun, M. Li, and G. Kuang, "Multimodal bilinear fusion network with second-order attention-based channel selection for land cover classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1011–1026, 2020.

[5] C. Qiu, M. Schmitt, L. Mou, P. Ghamisi, and X. Zhu, "Feature importance analysis for local climate zone classification using a residual convolutional neural network with multi-source datasets," *Remote Sens.*, vol. 10, no. 10, p. 1572, Oct. 2018.

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[6] N. Yokoya, P. Ghamisi, and J. Xia, "Multimodal, multitemporal, and multisource global data fusion for local climate zones classification based on ensemble learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 1197–1200.

[7] M. Kampffmeyer, A.-B. Salberg, and R. Jenssen, "Urban land cover classification with missing data modalities using deep convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 6, pp. 1758–1768, Jun. 2018.

[8] S. Aksoy, K. Koperski, C. Tusk, and G. Marchisio, "Land cover classification with multi-sensor fusion of partly missing data," *Photogramm. Eng. Remote Sens.*, vol. 75, no. 5, pp. 577–593, May 2009.

[9] R. Bahmanyar, D. Espinoza-Molina, and M. Datcu, "Multisensor Earth observation image classification based on a multimodal latent Dirichlet allocation model," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 3, pp. 459–463, Mar. 2018.

[10] P. J. García-Laencina, J.-L. Sancho-Gómez, and A. R. Figueiras-Vidal, "Pattern classification with missing data: A review," *Neural Comput. Appl.*, vol. 19, no. 2, pp. 263–282, Mar. 2010.

[11] M. Das and S. K. Ghosh, "A deep-learning-based forecasting ensemble to predict missing data for remote sensing analysis," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 12, pp. 5228–5236, Dec. 2017.

[12] L. Malan, C. M. Smuts, J. Baumgartner, and C. Ricci, "Missing data imputation via the expectation-maximization algorithm can improve principal component analysis aimed at deriving biomarker profiles and dietary patterns," *Nutrition Res.*, vol. 75, pp. 67–76, Mar. 2020.

[13] P. Khosravi, A. Vergari, Y. Choi, Y. Liang, and G. Van den Broeck, "Handling missing data in decision trees: A probabilistic approach," 2020, *arXiv:2006.16341*. [Online]. Available: http://arxiv.org/abs/2006.16341

[14] A.-B. Salberg and R. Jenssen, "Land-cover classification of partly missing data using support vector machines," *Int. J. Remote Sens.*, vol. 33, no. 14, pp. 4471–4481, Jul. 2012.

[15] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[16] L. Mou and X. X. Zhu, "Learning to pay attention on spectral domain: A spectral attention module-based convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 110–122, Jan. 2020.

[17] M. Kampffmeyer, A.-B. Salberg, and R. Jenssen, "Semantic segmentation of small objects and modeling of uncertainty in urban remote sensing images using deep convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Las Vegas, NV, USA, Jun. 2016, pp. 680–688.

[18] Y. Ren, C. Zhu, and S. Xiao, "Small object detection in optical remote sensing images via modified faster R-CNN," *Appl. Sci.*, vol. 8, no. 5, p. 813, May 2018.

[19] N. Jaques, S. Taylor, A. Sano, and R. Picard, "Multimodal autoencoder: A deep learning approach to filling in missing sensor data and enabling better mood prediction," in *Proc. 7th Int. Conf. Affect. Comput. Intell. Interact. (ACII)*, San Antonio, TX, USA, Oct. 2017, pp. 202–208.

[20] F. Pahde, M. Nabi, T. Klein, and P. Jahnichen, "Discriminative hallucination for multi-modal few-shot learning," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Athens, Greece, Oct. 2018, pp. 156–160.

[21] V. Vapnik, *Estimation of Dependences Based on Empirical Data*. Berlin, Germany: Springer, 2006.

[22] J. Hoffman, S. Gupta, and T. Darrell, "Learning with side information through modality hallucination," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 826–834.

[23] D. Lopez-Paz, L. Bottou, B. Schölkopf, and V. Vapnik, "Unifying distillation and privileged information," 2015, *arXiv:1511.03643*. [Online]. Available: http://arxiv.org/abs/1511.03643

[24] C. Choi, S. Kim, and K. Ramani, "Learning hand articulations by hallucinating heat distribution," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 3104–3113.

[25] F. Tang, C. Xiao, F. Wang, J. Zhou, and L.-W.-H. Lehman, "Retaining privileged information for multi-task learning," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Jul. 2019, pp. 1369–1377.

[26] N. Passalis and A. Tefas, "Learning deep representations with probabilistic knowledge transfer," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 268–284.

[27] E. Tzeng, J. Hoffman, T. Darrell, and K. Saenko, "Simultaneous deep transfer across domains and tasks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Santiago, Chile, Dec. 2015, pp. 4068–4076.

[28] L. Yu, V. O. Yazici, X. Liu, J. van de Weijer, Y. Cheng, and A. Ramisa, "Learning metrics from teachers: Compact networks for image embedding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 2902–2911.

[29] Y. Liu, C. Shu, J. Wang, and C. Shen, "Structured knowledge distillation for dense prediction," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, Jun. 12, 2020, doi: 10.1109/TPAMI.2020.3001940.

[30] N. Passalis, M. Tzelepi, and A. Tefas, "Heterogeneous knowledge distillation using information flow modeling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 2336–2345.

[31] S. Zagoruyko and N. Komodakis, "Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer," 2016, *arXiv:1612.03928*. [Online]. Available: http://arxiv.org/abs/1612.03928

[32] A. Romero, N. Ballas, S. Ebrahimi Kahou, A. Chassang, C. Gatta, and Y. Bengio, "FitNets: Hints for thin deep nets," 2014, *arXiv:1412.6550*. [Online]. Available: http://arxiv.org/abs/1412.6550

[33] Z. Ren, B. Hou, Z. Wen, and L. Jiao, "Patch-sorted deep feature learning for high resolution SAR image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 9, pp. 3113–3126, Sep. 2018.

[34] X. Jin *et al.*, "Knowledge distillation via route constrained optimization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Seoul, South Korea, Oct. 2019, pp. 1345–1354.

[35] Y. Bengio, J. Louradour, R. Collobert, and J. Weston, "Curriculum learning," in *Proc. 26th Annu. Int. Conf. Mach. Learn.*, 2009, pp. 41–48.

[36] I. Czarnowski, "Cluster-based instance selection for machine classification," *Knowl. Inf. Syst.*, vol. 30, no. 1, pp. 113–133, Jan. 2012.

[37] J. R. Cano, F. Herrera, and M. Lozano, "On the combination of evolutionary algorithms and stratified strategies for training set selection in data mining," *Appl. Soft Comput.*, vol. 6, no. 3, pp. 323–332, Mar. 2006.

[38] D. R. Wilson and T. R. Martinez, "An integrated instance-based learning algorithm," *Comput. Intell.*, vol. 16, no. 1, pp. 1–28, Feb. 2000.

[39] I. Czarnowski and P. Jędrzejowicz, "An approach to instance reduction in supervised learning," in *Proc. Int. Conf. Innov. Techn. Appl. Artif. Intell.*, 2003, pp. 267–280.

[40] C.-L. Chang, "Finding prototypes for nearest neighbor classifiers," *IEEE Trans. Comput.*, vol. C-23, no. 11, pp. 1179–1184, Nov. 1974.

[41] R. B. Anderson, J. F. Bell, III, R. C. Wiens, R. V. Morris, and S. M. Clegg, "Clustering and training set selection methods for improving the accuracy of quantitative laser induced breakdown spectroscopy," *Spectr. Acta B, At. Spectrosc.*, vol. 70, pp. 24–32. Apr. 2012.

[42] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*. [Online]. Available: http://arxiv.org/abs/1503.02531

[43] Z. Tang, D. Wang, and Z. Zhang, "Recurrent neural network training with dark knowledge transfer," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Shanghai, China, Mar. 2016, pp. 5900–5904.

[44] Z. Deng, Q. Kong, and T. Murakami, "Towards efficient instance segmentation with hierarchical distillation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Seoul, South Korea, Oct. 2019, pp. 3243–3249.

[45] M. Liu, X. Chen, Y. Zhang, Y. Li, and J. M. Rehg, "Attention distillation for learning video representations," 2019, *arXiv:1904.03249*. [Online]. Available: http://arxiv.org/abs/1904.03249

[46] OpenStreetMap. (2017). *Planet Dump Retrieved From*. [Online]. Available: https://www.openstreetmap.org.

[47] N. Kussul, M. Lavreniuk, S. Skakun, and A. Shelestov, "Deep learning classification of land cover and crop types using remote sensing data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 778–782, May 2017.

[48] N. C. Garcia, P. Morerio, and V. Murino, "Modality distillation with multiple stream networks for action recognition," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 103–118.

[49] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[50] A. Vali, S. Comai, and M. Matteucci, "Deep learning for land use and land cover classification based on hyperspectral and multispectral Earth observation data: A review," *Remote Sens.*, vol. 12, no. 15, p. 2495, Aug. 2020.

[51] Q. Liu, M. Kampffmeyer, R. Jenssen, and A.-B. Salberg, "Dense dilated convolutions' merging network for land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6309–6320, Sep. 2020.

[52] Y. Xiang, F. Wang, and H. You, "OS-SIFT: A robust SIFT-like algorithm for high-resolution optical-to-SAR image registration in suburban areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3078–3090, Jun. 2018.

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